

# Fluid Dynamics of the Atmosphere and Ocean, ECMM719, 2017.

## Problem Set 2A (Assessed)

This problem set counts 10% toward your final mark. It is due on 22 March, 2017. All questions should be attempted and all have equal credit, and unless marked the subsections within questions are also scored evenly.

Partial credit will be given if the algebra is wrong but the method right, *provided you explain what you are trying to do*. Mathematical communication counts for 10% of the marks.

### 1. Vertically propagating Rossby waves

- (a) The quasigeostrophic potential vorticity equation, linearized about a constant eastward mean flow  $U$  is

$$\left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) q + \beta v = 0,$$

where

$$q = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right)$$

is the perturbation potential vorticity. Here,  $\psi$  is the perturbation geostrophic stream function,  $z$  is a vertical coordinate,  $f_0$ ,  $N$  and  $\beta > 0$  are constants.

Derive the dispersion relation

$$\omega = kU - \frac{k\beta}{k^2 + l^2 + m^2 (f_0^2/N^2)}.$$

What kind of waves are these?

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- (b) Calculate the vertical component of the group velocity for such waves.

{4}

- (c) The perturbation  $v$  and  $b$  components will also have wavelike behaviour

$$(v, b) = \text{Re} \left\{ (\hat{v}, \hat{b}) \exp [i (kx + ly + mz - \omega t)] \right\},$$

where  $b = f_0 \partial \psi / \partial z$ . The northward eddy buoyancy flux is

$$\overline{vb} \equiv \left( \frac{1}{L_\lambda} \right) \int_{L_\lambda} vb \, dx$$

where  $L_\lambda$  is one wavelength. Show that

$$\overline{vb} = Akm |\hat{\psi}|^2,$$

where  $A$  is a constant. What is it? Hence explain why upward wave propagation requires a poleward eddy buoyancy flux.

{5}

- (d) Now consider *stationary* waves. Show that upward wave propagation is possible only if  $U$  satisfies the inequality

$$\frac{\beta}{k^2 + l^2} > U > 0.$$

{4}

{20}

## 2. Ekman layers

Consider a layer of fluid of constant density at in the upper ocean that satisfies the *Ekman-layer* equations:

$$-fv = -\frac{\partial \phi}{\partial x} + \frac{\partial \tau_x}{\partial z}, \quad fu = -\frac{\partial \phi}{\partial y} + \frac{\partial \tau_y}{\partial z}. \quad (\text{Ek})$$

where  $\tau_x, \tau_y$  are components of the stress,  $\boldsymbol{\tau}$ , in the  $x$ - and  $y$ -directions and  $f = f_0 + \beta y$ . Assume that the pressure,  $\phi$ , is not a function of  $z$ , that the Ekman layer has some finite depth,  $H_E$ , below which the stress is zero, and that the vertical velocity is zero at the top of the ocean,  $z = 0$ , and at the bottom,  $z = -H_o$ .

- Show that the total ageostrophic transport induced by the stress is at right angles to the direction of the surface stress.
- By integrating the mass continuity equation over the entire depth of the ocean show that the vertically integrated geostrophic velocity is given by

$$\int_{-H_o}^0 \beta v_g dz = f \left[ \frac{\partial}{\partial x} \left( \frac{\tau_{y0}}{f} \right) - \frac{\partial}{\partial y} \left( \frac{\tau_{x0}}{f} \right) \right]. \quad (\text{A})$$

- By cross-differentiating equations (Ek) and vertically integrating over the total depth of the ocean, or otherwise, derive the Sverdrup relation,

$$\int_{-H_o}^0 \beta v dz = \frac{\partial \tau_{y0}}{\partial x} - \frac{\partial \tau_{x0}}{\partial y} \quad (\text{B})$$

where  $v$  is the meridional component of the total velocity.

- If  $f$  is not constant eq. (A) is different from eq. (B). Explain how and why the two equations are nevertheless consistent. {20}

## 3. Rossby waves and jets

- Show that if there is a source of Rossby waves at any given latitude in the Northern Hemisphere, we expect that eastward flow will be generated there. Your answer will involve relating flux of momentum in Rossby waves and relating it to group velocity. How does your answer differ in the Southern Hemisphere?
- Consider two interacting Rossby waves in a single-layer barotropic fluid. Each Rossby wave generates a wave with a velocity amplitude of  $10 \text{ m s}^{-1}$ . Being explicit about the assumptions you make, what is the acceleration of the mean flow? How long will it take to generate a mean flow of  $20 \text{ m s}^{-1}$ ? {20}

#### 4. Geostrophic adjustment with a velocity jump.

- (a) Show that the *linearized* potential vorticity,  $q'$ , for the shallow water system is given by

$$q' = \zeta' - f_0 \frac{h'}{H},$$

using standard notation.

{5}

- (b) If the flow is in geostrophic balance show that the relative vorticity is given by

$$\zeta' = \nabla^2 \psi,$$

where  $\psi = gh'/f_0$ . Hence show that the potential vorticity is then given by

$$q' = \nabla^2 \psi - \frac{1}{L_d^2} \psi,$$

and write down an expression for  $L_d$ . (If you wish you may take  $g = 1$ .)

{5}

- (c) Suppose that the initial flow has a flat surface with  $u' = 0$  everywhere and Suppose that initially the fluid surface is flat, the zonal velocity ( $u$ ) is zero and the meridional velocity is given by

$$v(x) = v_0 \text{sgn}(x)$$

- (i) Find the equilibrium height and velocity fields at  $t = \infty$ , in the linear approximation.

- (ii) What are the initial and final kinetic and potential energies?

*Hint:* The potential vorticity is  $q = \zeta - f_0 \eta/H$ , so that the initial potential vorticity is given by

$$q = 2v_0 \delta(x).$$

where  $\delta(x)$  is the Dirac delta function.

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#### 5. Western boundary layers

Consider the barotropic vorticity equation in the form

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) + \beta v = -r\zeta + \nu \nabla^2 \zeta + F(x, y),$$

where  $\zeta = \nabla^2 \psi$ ,  $v = \partial \psi / \partial x$  and  $r$  and  $\nu$  are constants, and the flow is two-dimensional. We suppose the fluid is contained in a square container of side  $a$ , with  $0 \leq x \leq a$  and  $0 \leq y \leq a$  and  $F = -A \sin \pi y/a$  where  $A$  is a constant. We expect that the nonlinear term and both frictional terms are ‘small’, and we are interested in steady states for which  $\partial \zeta / \partial t = 0$ .

- (a) Nondimensionalize these equations, and obtain estimates of the sizes of each term. State explicitly the conditions under which each of the frictional terms, and the nonlinear term, are indeed small.
- (b) Neglecting the nonlinear term and both frictional terms, obtain the solution to  $\beta \partial \psi / \partial x = -A \sin \pi y/a$ . Can this solution satisfy the boundary conditions needed if the frictional terms are present. Explain briefly.
- (c) Suppose that  $\nu = 0$ , and neglect the nonlinear term, and assume that the term  $r\zeta$  is indeed small but nonzero. Show that we can expect a boundary current on one side of the ocean (which?) and estimate its thickness.
- (d) Now suppose that  $r = 0$ , and neglect the nonlinear term. Estimate the size of the boundary current that now arises.

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